

Discrete ORIGIN framework

Here we describe the discrete time version of ORIGIN used to generate the mitochondrial lineage results. In the associated work, mitochondrial parameterization operated under a restricted configuration of the framework described below. Namely, the following restrictions were imposed to narrow the scope of the research question to how mitochondrial inheritance patterns power lineage information:

- Cell cycle lengths were uniform within and across cell types
- Induced and uninduced clade partitions were established at the first cell division
- Cell death processes (stochastic, heteroplasmy-driven) were disabled to restrict analyses to complete binary trees
- Heteroplasmy variants were established at the founder level and not as downstream de novo processes

These restrictions can be relaxed according to the tunable factors described below.

Population initialization

Discrete ORIGIN initializes a cell population $\mathcal{C}(0) = \{c_0\}$ at time $t = 0$. Each cell c in $\mathcal{C}(t)$ is characterized by the state vector $c = (M_{mt}(c), M_{bc}(c), \tau(c), \theta(c))$, where:

- $M_{mt}(c)$ is the sparse mitochondrial mutation matrix of a cell c , a $g \times \ell_{mt}$ matrix, with g mitochondrial genome copies of length ℓ_{mt} each,
- $M_{bc}(c)$ is the sparse barcode mutation matrix of a cell c , an $m \times \ell_{bc}$ matrix, with a maximum of m barcode integrations of length ℓ_{bc} each,
- $\tau(c)$ denotes the cell type of cell c , and
- $\theta(c)$ denotes the cell- and cell type-specific parameters of cell c .

For analyses restricted to only mitochondrial data, $M_{bc}(c)$ is static and does not change over the course of the simulation.

25 Let n_{mito} denote the number of mitochondria in the founder cell. Define a map φ that
 26 assigns each genome row i of the $g \times \ell_{mt}$ mutation matrix $M_{mt}(c)$ to a mitochondrion
 27 $\varphi(i)$, where φ maps $\{1, \dots, g\}$ to $\{1, \dots, n_{mito}\}$. Let the inverse mapping from mitochondria
 28 to genome row indices be represented by φ^{-1} .

29 Let K be the number of distinct cell types. The discrete cell type $\tau(c)$ of a cell is defined
 30 when the cell arises from a division and may change to any other cell type at its own
 31 division point according to the $K \times K$ transition matrix Φ_0 . Binary inducible cell
 32 differentiation conditions can be represented by providing an induction timepoint t_I such
 33 that all cell type transitions in induced sublineages at timepoints $t \geq t_I$ use the $K \times K$
 34 transition matrix Φ_1 , where each row of Φ_r sums to 1 ($r = 0$ or 1). For a cell c born at
 35 $t_{c,birth}$ with type $\tau(c) = j$, its cell cycle length is parameterized by μ_j , where μ_j here is
 36 the fixed cell cycle length for cell type j . In the associated work, all cell cycle lengths
 37 across cell types are set to 1. Rather than specifying an induction timepoint, both Φ_0
 38 and Φ_1 exist concurrently from the first division; induction status is assigned at the first
 39 division of the founder cell such that it purely splits the lineage into two distinct
 40 sublineages. With zero cell death and perfect cell recovery, this means half the terminal
 41 population divides according to the induced regime and half to the uninduced regime.

42 While not used in these analyses, μ_j can optionally parameterize an exponential
 43 distribution to enable within-type cell cycle length heterogeneity. The next eligible
 44 division timepoint for this cell, $t_{c,elig}$, is defined as $t_{c,birth} + \mu_{\tau(c)}$. The simulation
 45 progresses from $t = 0$ to t_{sim} with discrete time points separated by a time increment
 46 Δt .

47 At each t , up to three processes can occur for each cell. For those cells past their
 48 eligible division timepoint, cell division occurs; stochastic and heteroplasmy-driven
 49 death events may prune the population; and all cells alive at t may undergo mutation
 50 processes that update M_{mt} and M_{bc} . During a cell division, daughter cells receive
 51 identical copies of parental barcode mutation matrices but perturbed copies of parental
 52 mitochondrial mutation matrices. Stochastic death processes occur with probabilities

$p_{death}(c)$ that are defined by $\tau(c)$. Transmission of mitochondrial profiles, mutation profile updates, and heteroplasmy-driven death are described in more detail below.

The output of the simulation is a discrete cell type-labeled heterogeneous cell population whose mutation matrices can be further processed to infer lineage relationships. By performing multiple post-hoc processing steps on the same mutation matrices, we can more easily draw comparisons between tree reconstruction performance metrics across a range of possible experimental limitations.

Mutational process encoding

Mutation matrices M_{mt} and M_{bc} store numeric representations of site- and integration-/genome-specific deviations from wild-type or baseline sequences across the lineage barcodes of a cell. We assume all barcode target sites are unedited at time $t = 0$, such that $M_{bc}(c)$ is the all-zero $m \times \ell_{bc}$ matrix; initial mitochondrial heteroplasmy makes this untrue with mitochondrial barcodes. For efficient representation of mutation profiles, we map nucleotides (A, G, C, T) to integers 1, 2, 3, and 4 respectively.

At each t , the barcode mutational matrices of all living cells undergo uniform and targeted mutational processes in two separate steps; the mitochondrial mutational matrices undergo only the uniform mutational process, which itself may be disabled such that mitochondrial variation is founder-derived only. Targeted mutational processes differ from their uniform counterparts by iteratively mutating barcode positions that have been predefined as targets and labeled with a discretized High, Medium, or Low edit rate classification to capture the variability in editing efficacy across guide/target combinations. The placement of target sites across the genomic barcode can be randomized or manually specified. Further heterogeneity within edit rate class is introduced by drawing from different quantile-defined regions of the underlying gamma distribution that specifies all target mutation probabilities of a given cell type and mutation type. These mutation types are defined as transitions, transversions, insertions, and deletions. In all mutation processes, we assume that multiple mutations cannot occur at the exact same base position in the same integration or genome, and we allow mutation rates to vary by cell type.

More formally, for each genomic position j and for each mutation type v , we generate the number of new integrations or genomes mutated, $E_{j,v}$, as a binomial draw with $N_j(\text{unedited})$ trials and success probability $\pi_{j,v}$, where $N_j(\text{unedited})$ is the number of unedited integrations or genomes at position j and $\pi_{j,v}$ is the probability of an edit of type v occurring in any one genome or integration at j . Note that in uniform editing processes, $\pi_{j,v} = \pi_v$ (independent of position). For each v , the $E_{j,v}$ integrations or genomes to be edited are then randomly sampled from the unedited pool at position j .

Inducible editing systems can be modeled in a binary way where prior to induction, all targets and non-targets alike undergo only background mutational processes; after induction, targets selectively undergo an additional round of non-uniform editing at each simulation timepoint.

For each cell at each simulation step, newly acquired mutations are encoded in temporary timepoint-specific mutation matrices $R_{mt}(c_t)$ and $R_{bc}(c_t)$. Without loss of generality in the mitochondrial context, a point mutation resulting in nucleotide base B that occurs in integration i at position j is encoded as $R_{mt}(c_t)[i, j] =$ (integer code for B). Insertions of length L_{insert} in integration i that begin at position j are encoded as fractional values less than 1, of the form $0.d_1d_2 \dots L_{insert}$, where each digit d_ℓ is the integer code for nucleotide B at that position of the insertion. Deletions of length L_{del} that start at position $j - L_{del} + 1$ in integration i are represented by the value -1 in each of the L_{del} consecutive entries from position $j - L_{del} + 1$ through position j .

Let $M_{bc}(c_t)$ be the incoming barcode mutation matrix, and let $M_{bc}(c_{t+\Delta t})$ be the barcode mutation matrix following these mutational processes, generated by

$$M_{bc}(c_{t+\Delta t}) = M_{bc}(c_t) + R_{bc}(c_t)$$

(similarly, $M_{mt}(c_{t+\Delta t}) = M_{mt}(c_t) + R_{mt}(c_t)$). This algebraic step ensures that mutations accumulated at time t are efficiently encoded into mutational profiles that can again be modified at future time steps.

Parameterization of CRISPR/Cas prospective lineage tracing modalities

Each barcode integration is assumed to have an identifiable tag that enables integrations to be compared across cells that descend from the same founder. For each specified target in a CRISPR-Cas system, editing windows may be built from optionally decaying edit rates with increasing distance from target sites. By indicating the size and behavior of this window, the user can allow for or prevent multiple edits occurring within the same target region.

Conventional Cas9 systems are encoded with a decaying editing window surrounding each target, as well as elevated insertion and deletion rates. Upon mutation of a target site within this editing window, further modifications are prohibited. To mimic the observed behavior of inter-target dropout that results from simultaneous deletions occurring at nearby genomic regions, we encode a dropout radius that probabilistically deletes all intervening genomic sequences between two sites when simultaneous deletions occur.

Base editing systems are modeled by increasing the frequency of either transitions or transversions at target sites, depending on the specified nucleotide conversion pattern. Target windows in a base editing context identify off-target nucleotides with the same identity as the target and encode them as possible sites of non-uniform editing.

In prime editing systems, insertion probabilities are increased with default editing windows. Deletion probabilities are decreased relative to nuclease systems to reflect the increased precision of the single-strand nick mechanism over double stranded breaks. Insertion sequences are drawn from a fixed library.

Mitochondrial dynamics

Unlike prospective barcodes that begin as a homogeneous population of unedited matrices, mitochondrial genomes exhibit underlying heterogeneity in the founder cell. We allow for a user-provided fraction of mitochondrial genome positions to exhibit some degree of heteroplasmy across all genomes in the founder cell. For simplicity, we assume that mitochondrial variants exist as single-base point mutations. To account for the potential impact of heteroplasmy on cellular fitness, we associate each variant with

137 a severity score drawn from a two-component Gaussian mixture density, with
 138 component means $\mu_{del} = -1$ and $\mu_{pro} = 1$ (arbitrary means corresponding to
 139 deleterious and protective scores), a shared standard deviation σ provided by the user
 140 that controls the shape of the density curve between μ_{del} and μ_{pro} , and a mixing weight
 141 (also user-provided) representing the fraction of variants that are negatively associated
 142 with cellular fitness.

143 We compute a cell-level heteroplasmy score across u_{mt} unique position-by-mutation
 144 combinations in the genome, where f_i is the fraction of mitochondrial genomes in the
 145 cell carrying mutation i and s_i is that mutation's severity score:

$$146 \quad S_c = \sum f_i \cdot s_i$$

147 At each cell division point, we allow mitochondrial fusion and fission events to occur. We
 148 simplify dynamics such that any two mitochondria within a cell can merge their genomes
 149 into one heritable unit, and any one mitochondrion can split its genomes into two
 150 mitochondria. Formally, we update our previously defined mitochondria-to-
 151 mitochondrial-genome map φ^{-1} :

152 During a fusion event, for two fusing mitochondria a and b joining to form mitochondrion
 153 c :

- 154 • $\varphi^{-1}(c) = \varphi^{-1}(a) \cup \varphi^{-1}(b)$
- 155 • $\varphi^{-1}(m)$ is unchanged for all m other than a , b , and c
- 156 • the new mitochondrial count $n'_{mito} = n_{mito} - 1$

157 During a splitting event of mitochondrion m into genome index sets I_m^1 and I_m^2 (whose
 158 union is I_m):

- 159 • $\varphi^{-1}(m_1) = I_m^1$
- 160 • $\varphi^{-1}(m_2) = I_m^2$
- 161 • $\varphi^{-1}(r)$ is unchanged for all r other than m_1 and m_2
- 162 • the new mitochondrial count $n'_{mito} = n_{mito} + 1$
- 163 • and $\varphi^{-1}(m)$ is removed.

Mitochondrial inheritance patterns

All mitochondria are replicated prior to each cell division point. Then, given the updated φ^{-1} , mitochondrial genomes are partitioned into daughter cells according to one of four mitochondrial inheritance mechanisms:

- Random: doubled mitochondrial pool is partitioned binomially between daughters without bias
- Biased: binomial partitioning of doubled mitochondrial pool with bias parameter $p_d \neq 0.5$. Daughter cell mitochondrial populations expand or constrict back to the parental number of mitochondria prior to downstream division events.
- Bottleneck: each daughter cell independently samples k_{bn} founder mitochondria from the parental pool, such that smaller values of k_{bn} correspond to larger drift per division. Expansion and/or constriction then preserve parental count.
- Clustered: mitochondria segregate as fused network clusters rather than individually, with a specified number of network clusters per cell that persist across divisions according to encoded persistence probabilities. Following a division event in which each daughter receives half of the parental clusters, the daughter clusters iteratively split according to size until the parental number of network clusters is restored.

Mitochondrial inheritance models are allowed to be cell-type specific. Inheritance regimes were defined by crossing unique combinations of inheritance patterns and fusion/fission parameters with cell types. The four inheritance regimes used in these analyses were chosen to reflect some of the most extreme edge cases under which mitochondrial signal could inform clonal structure.

Following this segregation, we transform cell-specific heteroplasmy scores into heteroplasmy-driven cell death probabilities. We formulate the probability that a cell c with heteroplasmy score S_c survives a heteroplasmy checkpoint during cell division using a logistic model:

$$P(\text{survive} | S_c) = \frac{1}{1 + \exp \left[- \left(\log \left(\frac{p_0}{1 - p_0} \right) + \beta (S_c - S_0) \right) \right]}$$

where S_0 is the heteroplasmy score of the founding cell, p_0 is the default survival probability of the founder, and we fix $\beta = 1$ such that more positive heteroplasmy scores are associated with increased survival probabilities.

We assume selection by heteroplasmy operates only at the cellular level rather than at the mitochondrial level and do not allow for the selective proliferation of lower-heteroplasmy mitochondrial genomes within a mitochondrion. We do permit the probabilistic loss of some fraction of mitochondria at each cell division point, independent of heteroplasmy, but this was disabled in the analyses presented here.

Post-mutational processing and lineage tree reconstruction

At the conclusion of the cell population simulation process, we allow for the selective downsampling of recovered terminal cells to provided cell type-specific recovery probabilities. Then, according to barcode integration and mitochondrial recovery probabilities, we further subset M_{bc} and M_{mt} , respectively, for each recovered terminal cell. During the mitochondria recovery process, we use φ_c^{-1} to extract the respective genome indices for each recovered mitochondrion. Note that we assume these recovery rates are uniform across integrations and mitochondria regardless of the underlying mutation burden. Thus, we assume missing mutational data is an uninformative feature in downstream tree building steps.

We use a look-up approach to map unique mutations to the cells containing those mutations. For barcode matrices, these unique mutations are identified by collapsing on integration number, genomic position, and mutation identity; for mitochondrial matrices, only genomic position and mutation identity are used to identify unique mutations, and allelic fractions for each variant at each genomic position are calculated. For a population with n_{term} terminal cells and u_{mt} and u_{bc} unique mitochondrial and barcode mutation combinations, we define two score matrices:

- W_{mt} : an $n_{term} \times u_{mt}$ matrix with entries in $[0,1]$ (continuous allelic-fraction scores for the mitochondrial mutations), and
- W_{bc} : an $n_{term} \times u_{bc}$ matrix with binary (0/1) entries for the barcode mutations.

If binary rather than allelic fraction analyses are desired, for a given allelic fraction threshold f_{af} , we further process the mitochondrial score matrix into a binary matrix by setting entry (c, u) to 1 if $A_{c,u} > f_{af}$, and 0 otherwise.

Distance-based mitochondrial lineage reconstruction

Unless otherwise noted, mitochondrial lineage trees are reconstructed with a pairwise distance-based clustering approach. For two cells indexed by A and B, the distance d between $W_{mt}[A,]$ and $W_{mt}[B,]$ is given by a form of Cavalli-Sforza chord distance (Cavalli-Sforza and Edwards 1967):

$$d(A, B) = \sqrt{L - \sum_l BC(l)}$$

where $BC(l) = \sqrt{W_{mt}[A, l]W_{mt}[B, l]} + \sqrt{(1 - W_{mt}[A, l])(1 - W_{mt}[B, l])}$ is the Bhattacharyya coefficient at variant position l and $L=u_{mt}$ is the number of position-by-variant combinations, where variants at a shared position are not independent. Trees are built from the generated distance matrix by neighbor-joining and rooting at the midpoint of the longest leaf-to-leaf path.

Clade definition approaches

Unless otherwise noted, mitochondrial mutation profile clades (MMPCs) are defined from the continuous allelic fraction matrix by iteratively grouping cells and merging clusters using the above chord distance and average-linkage clustering until there are at least 20 cells per clade (varied in Supplementary Results). Ground truth topological clades (GTTCs) are defined by partitioning the ground truth lineage tree into five (default) or 20 contiguous clades of terminal cells. Alternative clade definition schemes include Topo k (topological assignment of k clades, generalized to reconstructed trees rather than only ground truth trees as in GTTCs); Binary signature (aggregating cells

243 using a binary present/absent allelic signature matrix rather than an allelic fraction
244 matrix); and MMPC fixed k (average-linkage clustering of chord distances computed the
245 allelic fraction matrix into exactly k clades).

246 **Expectation maximization (EM) approach to classifying cell type transition regime**
247 **from terminal leaf states**

248 To illustrate the information capacity of mitochondria-derived clade assignments, we
249 develop a tree model with observed node states and latent transition process
250 assignments. Unlike a standard hidden Markov tree model, the cell states themselves
251 are treated as observed at the leaves, and the latent variables instead determine which
252 of two transition candidate matrices, Φ_0 and Φ_1 (uninduced and induced, respectively),
253 generated each internal subtree.

254 For each internal node v , define subtree X_v as the subtree rooted at v and containing all
255 descendants of v , and introduce a latent binary variable Z_v , where $Z_v = 0$ if X_v evolved
256 under Φ_0 , and $Z_v = 1$ if X_v evolved under Φ_1 .

257 Each clade c has a mixture parameter $p_c = P(Z_u = 1 \mid u \in c)$ that reflects the probability
258 that the divisions in clade c use the induced transition matrix Φ_1 . The EM procedure
259 iteratively seeks to estimate each p_c while keeping Φ_0 and Φ_1 fixed and defined a priori.

260 We assume that each leaf node v has an observed cell state x_k (one of K possible
261 states). We probabilistically encode these leaf cell types by defining a one-hot likelihood
262 vector for each leaf v : the i -th entry L_v^i equals 1 if $i = x_k$, and 0 otherwise.

263 For an internal node u with child v , the child likelihood vector is propagated upward
264 through each transition matrix. For process r (0 or 1), the propagated likelihood $M_r^v(i) =$
265 $P(X_v \mid x_u = i, \Phi_r)$, computed as the matrix product of Φ_r and the likelihood vector L_v .

266 We assume independence between children given a parent state. The subtree
267 likelihood for process r at node u is therefore the product over all children v of u , or
268 equivalently in log-space:

$$\log L_r^u(i) = \sum_{v \in ch(u)} \log M_r^v(i)$$

To prevent underflow, we subtract the maximum likelihood element of the uninduced transition matrix's log-likelihood vector from each entry.

Since subtrees rooted at each internal node may give rise to child nodes in multiple clades, we define a clade-weighted induced prior $\tilde{p}_v$ for each internal node v according to the relative representation of the downstream clades among the leaves of that subtree, representing $P(Z_v = 1)$. The subtree likelihood at node v is then modeled as a weighted mixture of the uninduced and induced likelihoods:

$$L_v(i) = \sum_{r \in \{0,1\}} P(Z_v = r) L_r^v(i) = (1 - \tilde{p}_v) \cdot L_0^v(i) + \tilde{p}_v \cdot L_1^v(i)$$

This likelihood vector is normalized by dividing by its maximum entry to obtain a normalized likelihood.

We fix the root cell state to be x_{root} and define its prior $\pi(i)$ to equal 1 when $i = x_{root}$, and 0 otherwise.

The total log-likelihood of all observed leaf states in the tree is

$$\log P(X) = \log\left(\sum_i \pi(i) \cdot L_{root}(i)\right) + \sum_{u \in V_{int}} \log factor_u$$

where $\log factor_u$ is the scaling correction term determined from the rescaling described above.

For each internal node v , EM finds the posterior probability $\gamma_v = P(Z_v = 1 | X)$. Given a parent prior vector q , the conditional likelihood under each induction scheme r is $P(X | Z_v = r) = A_r = \sum_i q(i) \cdot L_r^v(i)$, computed in log-space via the log-sum-exp of $\log L_r^v(i)$ weighted by $q(i)$.

Thus the posterior responsibility that node v contributes to determining whether the induced transition matrix Φ_1 was used is

$$\gamma_v = \frac{\tilde{p}_v \cdot A_1}{\tilde{p}_v \cdot A_1 + (1 - \tilde{p}_v) \cdot A_0}$$

Each node v is further assigned a tunable weight w_v to determine how strongly it contributes to the M-step. Uniform weighting was used for the analyses reported here.

For each clade c , we accumulate these weighted responsibilities and normalize by the total weight of all nodes that map to c :

$$N_c = \sum_{v \in c} w_v \cdot \gamma_v, \quad D_c = \sum_{v \in c} w_v$$

Under uniform weighting, $w_v = 1$ for every clade with a leaf beneath that node.

The induced fraction for each clade is then updated using a beta prior:

$$p_c = \frac{N_c + a_c - 1}{D_c + a_c + b_c - 2}$$

which is analogous to the MAP estimate of the beta distribution when set to $a_c = b_c = 1$. Informative clade priors may be derived according to factors like enrichment or rarity of cell types, but the analyses presented here used uninformative priors such that the update reduces exactly to the weighted maximum likelihood estimate. The model is initialized with an induced fraction of 0.5 for each clade and is run once rather than from multiple restarts, iterating until convergence or a maximum of 200 iterations before outputting clade-level induced probabilities. We threshold clade-specific induction probabilities at 0.5 and label all cells v in clade c with the induction regime y_v : “induced” if $p_c \geq 0.5$, and “uninduced” if $p_c < 0.5$. The implementation allows for post-hoc Gaussian mixture model transformations of outputted EM induction probabilities, but all probabilities used in these analyses use only raw EM probability outputs.

We apply this approach to a cell population that develops with two transition matrix regimes from the beginning, modeling hematopoietic stem cell development.

Evaluation Metrics

Normalized Robinson-Foulds distance was used to determine the reconstructive accuracy of lineage trees. Receiver Operator Curve – Area Under the Curve (ROC-AUC) was appropriate for evaluating the performance of the EM model because of its balanced binary classification nature. Adjusted mutual information (AMI) was used as the primary metric to quantify the concordance between sets of clades because of set size imbalances. Other clade-aware metrics considered were adjusted Rand index (ARI) and F1 purity (defined as the harmonic mean of two directional purities: in one direction, each MMPC is matched to the GTTC to which most of its cells belong and purity is defined as the fraction of cells in that MPPC that belong to that GTTC; the other direction reverses the roles of MMPC and GTTC). Of these clade-based metrics, AMI showed the lowest correlation with number of clades ($|\rho| = 0.32$) (**Figure S13**). Three clade-agnostic measures were also computed for comparison in **Figure S13**:

- Mantel correlation: Spearman correlation between cell-pairwise ground-truth patristic distances and allelic fraction chord distances
- Cophenetic correlation: same form as Mantel correlation, but replaces allelic fraction chord distances with patristic distances on the reconstructed mitochondrial tree (built with neighbor joining of chord distance allelic fraction matrix)
- Triplet consistency: samples 3000 random cell triplets, identifies which of the three pairs among the triplet is closest under ground-truth patristic distance and allelic fraction chord distances, and finds the fraction of triplets on which the two agree (chance = $\frac{1}{3}$).

Allele fraction variation was decomposed against GTTCs by iterating over mitochondrial genome positions and, for each, splitting total variance into a between-clade and within-clade component (**Figure S11A**). Between-clade variance was calculated by comparing clade mean to grand mean, and within-clade by looking only at the spread within each clade. Formally, for each position u , across n cells distributed across clades of size n_k :

$$\text{Var}_u = \frac{1}{n} \sum_c (x_{cu} - \bar{x}_u)^2 = \sum_k \frac{n_k}{n} (\bar{x}_{ku} - \bar{x}_u)^2 + \sum_k \frac{n_k}{n} \text{Var}_{ku}$$

where the two summed components are the between-clade and within-clade variance components, respectively.

Supervised ceiling of mitochondrial clades

The horizontal bars in Fig. 6.2B address whether clades contain sufficient information to classify their induction states irrespective of whether this signal is capturable by the EM modeling approach. For each combination of simulation, timepoint, and clade definition, cells are randomly assigned to one of two halves, A and B. Each cell i in each partition (A or B) of each MMPC is assigned a probability of being induced by finding the fraction of cells in the other half (B if cell i is in A; A if cell i is in B) that are induced. A value near 1 means clades are close to homogeneous in induction status, while a value near 0.5 suggests balanced mixing of induction status that no classifier could capture. Of note, this quantity is supervised, whereas the EM mixture model must infer clade state from cell type transitions alone; it is thus a ceiling, and the difference between it and each achieved ROC-AUC is the signal present but not exploited.

Alpha sweep sensitivity analysis of input transition matrices

The induction inference workflow requires a priori knowledge of transition matrices for both induced and uninduced conditions, yet this information may not be exactly known in a real world context implementation. To determine the effects of uncertainty in the input transition matrices on EM performance, we add increasing noise to the induced transition matrix according to a mixture parameter α , such that

$$\Phi_1^*(\alpha) = (1 - \alpha)\Phi_0 + \alpha\Phi_1$$

This sensitivity test captures how the prior passed to the EM modulates model performance as α varies from 0 (fully uninformative: every row equal to Φ_0 ; every likelihood ratio is exactly 1) to 1 (maximal discriminability: every row equal to Φ_1). The α sweep was evaluated at $\alpha = 0, 0.1, 0.25, 0.5, 0.75, 1$ (**Figure S14**).

Model comparisons

In addition to the supervised ceiling described above, induction inference of the EM model was compared to the following tree- and/or model-free controls:

371 Cell type alone: all cells are assigned to one of two folds, and each cell in a given fold is
372 scored by the observed probability of being induced given its own cell type, estimated
373 using only cells in the opposite fold.

374 Crossed-pairing score: each clade is scored by the log ratio of counts of co-occurring
375 cell types that differ between Φ_0 and Φ_1 : $\log \frac{n_{MyMPP} * n_{Lym} + n_{LyMPP} * n_{Mye} + 1}{n_{MyMPP} * n_{Mye} + n_{LyMPP} * n_{Lym} + 1}$. The +1 is a
376 pseudocount.

377 Sibling-pair likelihood ratio: looks only at terminal sibling pairs of types a, b within one
378 clade, as these cells are each a single step away from the same shared parental type,
379 and computes $\log \frac{L_1}{L_0}$, where $L_r = \sum_t \pi_t * \Phi_r[t, a] * \Phi_r[t, b]$ under leaf-type distribution π .
380 This is a single pass with no pruning and no iteration. Clade scores are averaged log-
381 ratios.

382 **References**

383 Cavalli-Sforza, L.L., and A. W. F. Edwards. 1967. "Phylogenetic Analysis: Models and
384 Estimation Procedures." *Evolution* 21 (3): 550-570.